MTH636 – Game Theory

Lecture 9: Mixed Strategies

1 Motivation for Mixed Strategies

In many games, if players are restricted to pure, deterministic strategies, they might not be able to guarantee a high payoff, and the game might not have a Nash Equilibrium or a Value.

Example: Advantage of Randomization

Consider the following zero-sum game for Player 1:

$P_1 \setminus P_2$	L	R	**Min**
T	4	1	1
B	2	3	**2**

In pure strategies, Player 1's max-min value is $\underline{v}_1 = 2$ (by playing B).
What if Player 1 plays a **mixed strategy**? Suppose Player 1 plays T with probability 1/4 and B with probability 3/4.

- If P2 plays L, P1's expected payoff is: $\frac{1}{4}(4) + \frac{3}{4}(2) = 1 + 1.5 = 2.5$
- If P2 plays R, P1's expected payoff is: $\frac{1}{4}(1) + \frac{3}{4}(3) = 0.25 + 2.25 = 2.5$

By randomizing this way, Player 1 guarantees a payoff of 2.5, which is strictly greater than the pure max-min value of 2.

2 Formalizing Mixed Strategies

Definition: Mixed Strategy

Let $G = \langle N, (S_i)_{i \in N}, (u_i)_{i \in N} \rangle$ be a strategic form game in which the set of strategies S_i of all players is **finite**.
A **mixed strategy** of Player i is a probability distribution over their pure strategies S_i. The set of all mixed strategies of Player i is denoted by Σ_i:

$$\Sigma_i = \left\{ \sigma_i : S_i \to [0,1] \;\middle|\; \sum_{s_i \in S_i} \sigma_i(s_i) = 1 \right\}$$

If $S_i = \{T, B\}$, a mixed strategy can be written as $\sigma_1 = (p, 1-p)$, meaning play T with probability p and B with probability $1-p$.

Definition: Mixed Extension of a Game

The **mixed extension** of the game G is a new game:

$$\Gamma = \langle N, (\Sigma_i)_{i \in N}, (U_i)_{i \in N} \rangle$$

where Σ_i is the set of mixed strategies for Player i, and U_i is the **expected utility**

function, defined as:

$$U_i(\sigma_1, \sigma_2, \dots, \sigma_n) = \sum_{(s_1,\dots,s_n)\in S} u_i(s_1, \dots, s_n)\sigma_1(s_1)\sigma_2(s_2)\cdots\sigma_n(s_n)$$

3 Nash Equilibrium in Mixed Strategies

Definition: Mixed Strategy Nash Equilibrium

Every Nash Equilibrium of the mixed extension Γ is called a **Nash Equilibrium in mixed strategies** of the original game G.
A profile $\sigma^* = (\sigma_1^*, \dots, \sigma_n^*)$ is an NE in mixed strategies if:

$$U_i(\sigma^*) \geq U_i(\sigma_i, \sigma_{-i}^*) \quad \forall i \in N, \forall \sigma_i \in \Sigma_i$$

Theorem: Checking NE via Pure Deviations

A profile σ^* is a mixed strategy Nash Equilibrium if and only if no player can gain by deviating to a **pure strategy**:

$$U_i(\sigma^*) \geq U_i(s_i, \sigma_{-i}^*) \quad \forall i \in N, \forall s_i \in S_i$$

Proof Intuition: Consider any mixed deviation σ_i. The expected utility $U_i(\sigma_i, \sigma_{-i}^*)$ is just a weighted average (convex combination) of the utilities of the pure strategies:

$$U_i(\sigma_i, \sigma_{-i}^*) = \sum_{s_i\in S_i} \sigma_i(s_i)U_i(s_i, \sigma_{-i}^*)$$

If $U_i(\sigma^*) \geq U_i(s_i, \sigma_{-i}^*)$ for every pure strategy s_i, then any weighted average of these pure strategy payoffs will also be less than or equal to $U_i(\sigma^*)$.

$$\sum_{s_i\in S_i} \sigma_i(s_i)U_i(s_i, \sigma_{-i}^*) \leq \sum_{s_i\in S_i} \sigma_i(s_i)U_i(\sigma^*) = U_i(\sigma^*)\sum \sigma_i(s_i) = U_i(\sigma^*)$$

4 Linearity of Expected Utility

Expected utility is **multilinear** in the probabilities. If we hold all other players' mixed strategies fixed, the expected utility U_i is a linear function of Player j's mixed strategy σ_j:

$$U_i(\dots, \lambda\sigma_j + (1-\lambda)\sigma_j', \dots) = \lambda U_i(\dots, \sigma_j, \dots) + (1-\lambda)U_i(\dots, \sigma_j', \dots)$$

This linearity is the fundamental property that allows us to solve for mixed equilibria using indifference conditions (as we will see).